bioacoustic_pipeline

A **falsifiable, multi-view pipeline for bioacoustic structure discovery.**

This is not a classifier that emits labels. It decomposes vocal audio into

acoustic s

Audio

| Corpus

Design principles

1. **No single model produces "the answer."** Each module emits an auditable signal; s

Layout

src/bioacoustic_pipeline/

ingestion

Install (editable)

```
pip install -e ".[dev,ml]"          # core + torch/sklearn/onnx
```

```
pip install
```

Commands

Run the pipeline (audio → verdict)

```
python sc
```

Track A: engineering validation on LibriSpeech

```
python sc
```

Track B: bat (corrected, annotation-grounded)

python sc

Track B: marmoset (call-type labels)

python sc

Cross-species + representation-dependence

python sc

Determinism + reproducibility (Claim A)

python sc

Negative control (ESC-50 environmental sound)

python sc

Playback design (Phase 1: falsifiable stimuli + manifest)

python sc

Train species tokenizers

python sc

Cross-corpus DUSTED recurrent acoustic-term discovery

Marmoset (HECB sub-call units, oracle)

python scripts/hecb_dusted_marmoset.py --n-eval 1500

Bengalese finch (GT syllable-validated positive control)

python scripts/dusted_bengalese.py --birds 0 1 2

Egyptian fruit bat (bout-level, emitter vs context)

python scripts/dusted_bat.py --n-bouts 2000

Bat held-out emitter validation (70/30 split)

python scripts/bat_heldout.py --n-bouts 2000

RookID individual-ID (whole-call vs HECB, cross-session held-out)

python scripts/rookid_individual_id.py --max-calls 3000

RookID-DUSTED motif discovery

python scripts/rookid_dusted.py --max-calls 2000

Unsupervised word-structure benchmark (LibriSpeech)

§14: HECB vs HuBERT vs CPC vs fingerprint same-word consistency

python scripts/unsupervised_word_structure_benchmark.py --n-files 200

Stage C: DUSTED word-term discovery (DTW, HuBERT's native metric)

python scripts/dusted_librispeech_words.py --n-files 200

Stage D: ES-KMeans-style word-embedding clustering

python scripts/eskmeans_librispeech.py --n-files 200

§16: HuBERT sanity ablation (layer × k × collapse × pooling grid)

python scripts/hubert_ablation.py --n-files 200

Lint + test

ruff check

Architecture v1.0: boundary sweep + alternatives + taxonomy + playback + controls

§30: boundary-quality sweep (whole-call primary, meerkat + marmoset)

python scripts/boundary_quality_sweep.py --n-clips 300 --n-per-type 500

§33: DUSTED vs 4 structural alternatives (per-dataset regime)

python scripts/compare_structure_methods.py --n-files 20

§34: targeted reanalyses (bat RQA-by-context / marmoset MP / Bengalese network)

python scripts/bat_rqa_by_context.py --per-context 30 --top-k 6

python sc

Step 3: repertoire taxonomy + Step 4: identity-form-context

python scripts/repertoire_taxonomy.py

python sc

§36 Entanglement Ceiling — AVES ablation (reviewer-proofing: deep SSL control)

python scripts/entanglement_ceiling_aves.py --n-per-type 500 --min-calls-per-caller 20

§37 Compositional-structure mining (structural prerequisites T0 + T2 stimuli)

python scripts/compositional_candidates_bat.py --n-bouts 2000 --k 24

§38 Compositional scan — zebra finch call-type sequences (filename-only, instant)

python scripts/compositional_zebrafinch.py --age adult

§39 Cross-corpus RQA — graded recurrent units / motif substrate

python scripts/rqa_cross_corpora.py --n 80 --corpora bat,marmoset,zebra_finch,bengalese,librispeech

§40 Individual vs group recurrent-unit discovery (is "isolate individuals" always needed?)

python scripts/individual_vs_group_discovery.py --n-per-type 500 --min-calls-per-caller 20

§41 Attribute recurrent structures to within-individual vs across-group (Herfindahl vs null)

python scripts/recurrent_structure_attribution.py --age adult --min-count 5

§42 Same attribution across corpora (zebra finch / marmoset / Bengalese / LibriSpeech)

python scripts/recurrent_structure_attribution_cross_corpora.py

§43 RQA on RookID (per-call seek from 2h field wavs; completes the §39 table)

python scripts/rook_rqa.py --max-calls 400

Step 6: DSP Audio Splicer (5-condition counterfactual WAVs)

python scripts/playback_splice.py --n-bouts 20 --out outputs/playback_stimuli

§35: negative controls (synthetic chirps / random segments)

python scripts/extended_negative_controls.py --n-bouts 50

Step 11: stereotypy vs template (cross-caller retrieval)

python scripts/stereotypy_vs_template.py --n-per-type 200

Factorial: HECB typed vs scalar+CPC on LibriSpeech boundary F1

python scripts/hecb_calibrator_factorial.py --n-train 400 --n-test 100

Validated results

Track A (LibriSpeech, 150 files)

Metric	Result
§1 combined/multi-resolution F1 (CPC	**0.836** (P=0.77, R=0.92)
§2 Semantic ceiling (token→word consiste	0.6% — tokens ≠ word identity
§3 Linear probe (word identity from CPC	35.8% / **71× lift** over 199 classes
CPC latent → speaker probe	0.56 (speaker-dominated representation)

Track B: Bat (Egyptian fruit bat, 23k segments, within-Emitter null)

Representation	context_binding nmi	Cramér's V	p
Fingerprint (deterministic physics)	0.027	0.148	0.007
Spectrogram (deterministic, baseline	**0.138**	**0.197**	0.007
AVES (learned multi-taxa SSL)	0.030	0.162	0.007

Representation-ROBUST: 3/3 significant, within-Emitter-controlled. The

determini

Track B: Marmoset (common marmoset, 12.5k calls, 5 call types)

Representation	token↔call-type V	probe acc	p
Fingerprint (deterministic)	0.606	0.83	0.007
AVES (learned multi-taxa SSL)	**0.698**	**0.98**	0.007

Representation-ROBUST: 2/2 significant. Call types (Phee/Twitter/Trill/Tsik/Seep) recovered

Functional aliasing (representation-dependent)

Tokenization	human alias density	bat alias density
Per-species VQ-VAE	4%	0%
Shared fingerprint	0%	17.5%

Functional aliasing **flips sign** between representations → it is **not a species discrimina

Determinism (Claim A: implementation reproducibility)

Artifact	Run 1	Run 2	Run 3	Status
verdict.json	identical	identical	identical	
tokens.jsonl	identical	identical	identical	
motifs.json	identical	identical	identical	
(all 9 artifacts)	—	—	—	**all IDENTICAL**

Three independent subprocess runs under a fixed seed (1337), CPU, deterministic DSP (torc

Negative control: ESC-50 (BEANS-Zero subset)

Metric	Result
Final verdict	**NON_BIOACOUSTIC_COMMUNICATION**
Pipeline outputs	NBD segments + fingerprints + tokens **c
Biological axes	functional_aliasing, motif, turn, contex
Biological-source clips (dog, crickets,	`BIOLOGICAL_SOUND_NON_COMMUNICATION_BENC
Non-biological clips (rain, engine, …)	`NON_BIOLOGICAL_SOUND`

The pipeline detects acoustic structure (segments, fingerprints, tokens) but the domain g

Cross-corpus DUSTED recurrent acoustic-term discovery (§11–§13, §18–§19)

corpus	regime	inventory	key finding
Bengalese finch (§12)	discrete syllable song	diverse, high-purity	fixed-cosine wmean purity **0.895**; DTW
Egyptian fruit bat (§13)	graded bout vocalisations	graded/intermediate	Emitter V **~0.46** ≫ Context V **~0.16*
InfantMarmosetsVox (§11)	discrete calls (oracle HECB)	sparse, one dominant	DTW calltype V **0.51** = 4.7× exact; de
RookID (§19)	noisy outdoor corvid calls	degenerate (one dominant)	Event V **0.22** > Source V **0.08**; mo

The difference is the corpus/repertoire, not the method — DUSTED lands where predicted by vocaliz

Bat held-out emitter validation (§17)

metric	Train (70%)	Held-out (30%)
Emitter V	0.221	**0.375**

Context V	0.085	0.149
ratio	2.60×	**2.52×**

The emitter/caller claim **survives held-out** — not a session/file artifact.

RookID individual-ID (§18)

representation	Source V	macro-F1	held-out control
whole-call fp (44-D)	**0.384**	**0.287** (4× chance)	recording-date held-out
HECB-pooled fp (44-D)	0.321	0.263	(sub-call does NOT improve ID)

LibriSpeech unsupervised word-structure benchmark (§14–§16)

unit source	within/across ratio	NN acc	word-cluster V
fingerprint (44-D)	**0.964**	**0.147**	**0.517**
CPC-km(50)	0.954	0.191	0.436
HECB	0.976	0.089	0.392
HuBERT-km(50)	1.008 (fails)	0.052	0.338 (fails)

HuBERT is metric-dependent: fails Levenshtein/clustering but recovers under DTW

(Stage C:

Boundary-quality sweep — whole-call primary confirmed (§30)

Corpus	Parent F1	whole V	pooled V	gap (pooled−whole)	Verdict
MeerKAT (n=300, oracle)	1.000	0.364	0.386	+0.021	hierarchy barely helps
Marmoset (n=2500, ~~or~~acle)		0.524	0.492	**−0.032**	hierarchy HURTS

Sub-call internal structure is augmentation for motif discovery, NOT a call-type-recovery

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Multi-paradigm structural validation (§33–§34)

DUSTED corroborated by 4 independent mathematical paradigms:

Paradigm	Dataset	Key result	Verdict vs DUSTED
PST + Graph	Bengalese	247 contexts, 5 communities, cf. Sainburg	**Corroborates**
RQA	Bat	%LAM context-dependent (**p=0.016**), Δ=0.05	**Complementary** (dynamics DTW missed)
Matrix Profile	Marmoset	87.5% stutter-dominant, unimodal histogram	**Challenges** degenerate, token-free
Transition Network	LibriSpeech	639 nodes, 19 communities, path 1.257	**Upper bound** (scales)

Cross-species repertoire taxonomy (§31)

Corpus	Type	Discriminator
Bengalese finch	`discrete_syllable`	cov 0.48, null 2.67×, bootstrap 0.509
Egyptian fruit bat	`graded`	cov 0.15, emitter V 0.45, social_lift 1.
Infant marmoset	`sparse_repetitive`	cov 0.028, bootstrap 0.229
Rook	`individually_stereotyped`	Source V 0.31 ≫ social_other 0.04
LibriSpeech	`human_lexical`	terms are NOT words (semantic ceiling)

Identity–form–context disentanglement (§32)

Target	Bat	Marmoset	Rook	LibriSpeech
Individual ID	**strong** (V 0.45)	n/a	partial (V 0.31)	strong (probe 0.56)
Call/event type	weak (V 0.17)	**strong** (V 0.60, probe 0.83)	**strong** (V 0.67)	partial (token↔word 0.29)
Context	**needs social** (D 4.3×)	n/a	weak (V 0.04)	n/a
Social partner	n/a	n/a	n/a (≈chance)	n/a

Negative controls — pipeline specificity (§35)

Control	# motifs	vs natural (33)	Specificity
Synthetic FM chirps	**1**	33×	PASS
Random audio segments	**1**	33×	PASS

The pipeline does NOT manufacture structure from arbitrary input.

The Entanglement Ceiling — semantic ceiling is mathematical (§36)

Four methods tested to separate caller identity from call form in the 44-D fingerprint:

Method	Mathematics	Δ caller V	Δ form V	Result
LDA projection	Linear subspace	−0.029	**−0.120**	FAIL (overlap)
LogReg projection	Linear weight	−0.005	−0.013	FAIL (inseparable)
Z-normalization	Affine	−0.019	**−0.074**	FAIL (overlap)
Delta-trajectory	Non-linear	−0.105	**−0.163**	FAIL (temporal dynamics are caller-s

All four fail. Caller identity and call form are non-separably entangled at every level

(static spe

Learned-representation control (AVES, §36). The same linear disentanglement was repeated

on a 768-

Modules

Sequence-Sufficiency Gate (`diagnostics/sequence_sufficiency.py`)

Prevents f

Domain Gate (`diagnostics/domain_gate.py` + `negative_control_report.py`)

Runs **be

Acoustic Fingerprint (`diagnostics/acoustic_fingerprint/`)

Physics-g

Spectrogram Bridge (`representation/spectrograms/`)

Multi-view

Segment-Representation Source (`embeddings/segment_representation.py`)

Makes the

Context Binding (`diagnostics/context_binding/`)

Annotatio

Speaker-Controlled Alias Null (`alias_detector.controlled_alias_tests`)

Functiona

DUSTED Recurrent Subsequence Motifs (`motif_miner/subsequence_motifs.py`)

Adapts sp

Playback Design — Phase 1 (`experiments/playback_design/`)

Turns diag

Determinism + Reproducibility (`determinism.py` + `hashing.py`)

Frozen-inf

Ethological Firewall (`embeddings/firewall.py` + `configs/model_metadata.json`)

JSON-driv

Runtime Layer (`runtime/`)

Device m

H0-CMC Evidence Ladder (`diagnostics/h0_cmc/`)

Six-rung c

Structural Alternatives (`diagnostics/structural_alternatives/`)

Four firew

Repertoire Taxonomy (`diagnostics/repertoire_taxonomy/`)

Pure-stdli

DSP Audio Splicer (`experiments/playback_design/audio_splicer.py`)

Track C: g

Caller-Identity Disentanglement (`representation/disentangle.py`)

Three me

Channel Audit + Temporal Ecology (`diagnostics/`)

`channel_

Status

Track A (engineering): validated on LibriSpeech. §1 combined-F1 = 0.836.

Semanti

Session 2 — representation science + Track C freeze (§21–§29, 2026-06-23)

Unified LS word-structure (§21): same 600 words × 89 types for §14 + Stage C + D.

Finge
metric-de

New scripts: `ls_unified_rerun`, `ls_speaker_invariant`, `ls_consensus_template`,

`ls_aves_e

Session 3: Architecture v1.0 — multi-paradigm validation + Track C (2026-06-23)

Architecture v1.0 freeze (12 steps, all complete)

Router au

Multi-paradigm structural validation (§33–§34)

DUSTED c

H0-CMC evidence ladder + Track C

**Evider

New modules: `diagnostics/{h0_cmc, repertoire_taxonomy, structural_alternatives,

channel_a

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